

UK Online Retail Sales & Growth Analysis

Forage Data Analytics Job Simulation
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Executive Summary

This report presents an analysis of transactional sales data from a UK-based online retailer covering the 2010–2011 period. Exploratory data analysis was conducted using SQL to clean, structure, and interrogate the raw dataset, with findings visualized in an interactive Power BI dashboard. The objective was to identify revenue drivers, seasonal demand patterns, high-value customer segments, and international expansion opportunities to support strategic decision-making by senior leadership.

The business generated a total of **£10.65M** in revenue across **38 regions**, serving **6,528 customers**. The United Kingdom remains the dominant market, contributing **85% of total revenue**, while year-over-year sales growth stood at **11.92%**. Revenue was heavily concentrated in the final quarter of 2011, pointing to a strong and predictable seasonal buying pattern that offers a clear opportunity for more targeted forecasting and marketing investment.

Methodology

Analysis was carried out in two stages. First, SQL was used to perform exploratory data analysis on the raw transactional dataset — profiling record volumes, identifying and handling data quality issues (such as cancelled orders and missing customer IDs), and running aggregate queries to surface early patterns in revenue, product performance, and customer behavior. Second, the cleaned data was modeled in Power BI, where DAX measures and interactive visuals were built to translate these findings into a stakeholder-facing dashboard, tailored to the specific questions posed by CEO and CMO personas.

Key Findings

- **Seasonal revenue concentration:** Monthly revenue held relatively flat between £0.5M–£0.8M from January through September 2011, then climbed sharply to a peak of £1.50M in November. December 2011 shows a sharp drop, but this reflects an **incomplete month of data** rather than an actual decline in demand — the dataset cuts off partway through December. Had a full month of data been available, December would likely have ranked among the strongest months of the year, in line with the pre-holiday buying surge already visible in November.
- **UK market dominance:** The UK accounts for 85% of total revenue, confirming it as the core market, while £1.38M in revenue came from the top 10 countries outside the UK.
- **Leading international markets:** Excluding the UK, the Netherlands, EIRE (Ireland), and Germany generated the highest revenue, followed by France and Australia — indicating concentrated demand in Western Europe.
- **Top-performing products:** Best sellers by revenue included the Regency Cakestand 3 Tier, Paper Craft Little Birdie, and the White Hanging Heart T-Light Holder, with Jumbo Bag Red Retrospot and

Medium Ceramic Top Storage Jar leading by quantity sold.

- **High-value customers:** A small group of customers contributes disproportionately to revenue, with the top customer generating over £280K — highlighting a clear opportunity for targeted retention efforts.
- **Expansion potential:** Geographic demand mapping (excluding the UK) highlights clusters of activity across Western Europe and pockets of demand in North America and Australia, suggesting realistic near-term targets for international growth.

Recommendations

- **Plan for Q4 seasonality:** Align inventory, staffing, and marketing spend with the sharp October–November demand spike, and account for December's likely full-month potential (understated in this dataset) when forecasting for 2012.
- **Prioritize European expansion:** Invest in targeted marketing and localized logistics in the Netherlands, Ireland, and Germany, where demand is already established outside the UK.
- **Launch a loyalty program for top customers:** Protect and grow revenue from the highest-spending customer segment through personalized offers and retention campaigns.
- **Double down on proven bestsellers:** Ensure consistent stock availability for top-performing products such as the Regency Cakestand and White Hanging Heart T-Light Holder, which show strong demand across both revenue and volume.
- **Monitor emerging markets:** Track early demand signals in North America and Australia to evaluate them as second-phase expansion targets.

Conclusion

The data points to a business with a strong, stable UK core and clear, data-backed opportunities for controlled international growth. By combining SQL-driven exploratory analysis with an interactive Power BI dashboard, this project translated raw transactional data into insights that map directly onto strategic decisions around forecasting, customer retention, and market expansion.